Predicting and Explaining Customer Churn

Data & AI in Economics

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IBM Telco Customer Churn

Full project overview for the Data & AI in Economics mission

Research question:
Does contract type causally increase churn probability, and does this effect differ across customer segments?

7,043
raw customers

7,032
after cleaning

26.6%
observed churn

Causal Inference

What would happen under a contract or support intervention?

Supervised Learning

Who is likely to churn next?

Unsupervised Learning

Which customer types exist?

Data cleaning and EDA

The exploratory analysis motivates the modeling choices but does not make causal claims.

Cleaning decisions

TotalCharges converted to numeric; 11 blank rows removed. customerID dropped. Churn mapped to 0/1. Service labels standardized.

Class distribution

Key EDA findings

- Month-to-month customers churn much more than two-year customers.
- Tenure is central: churn is concentrated among newer customers.
- Higher monthly charges are associated with churn.
- TotalCharges and tenure are highly correlated, so interpretation must be careful.

Implication: use stratified evaluation and metrics beyond accuracy because the majority class is non-churn.

Causal inference: from correlation to intervention

DoWhy backdoor adjustment estimates effects under explicit DAG assumptions.

Primary treatment: Contract

ATE = -0.1305. Moving one step toward a longer contract is associated with about a 13 percentage-point reduction in churn after adjustment.

Alternative treatment: Tech Support

ATE = -0.1036. Providing technical support is estimated to reduce churn probability by about 10.4 percentage points after adjustment.

Refutation checks

✓ **Random common cause: estimates remain stable**
✓ **Placebo treatment: effect collapses near zero**

Main methodological boundary

These are observational causal estimates. They are useful for economic interpretation, but they depend on the DAG, observed confounders, and no-unmeasured-confounding assumptions.

Causal question: what intervention may reduce churn?

Prediction and segmentation

Supervised learning scores risk; K-Means identifies where risk is concentrated.

Supervised test-set results

Model	ROC	PR	Prec.	Rec.	F1-S.	Acc.
Dummy	0.500	0.266	0.000	0.000	0.000	0.734
Logistic	0.836	0.621	0.645	0.578	0.609	0.803
Forest	0.840	0.646	0.520	0.786	0.626	0.751

Recall priority: Random Forest

Precision priority: Logistic

K-Means segment profiles

Loyal Premium
N=2,344 · tenure ~59 · $75.27/mo · churn 9.0%

Transactional Volatiles
N=3,634 · tenure ~15 · $54.14/mo · churn 32.0%

High-Value At-Risk
N=1,054 · tenure ~30 · $78.27/mo · churn 45.0%

Integrated implication: High-Value At-Risk has the highest churn rate, while Transactional Volatiles is the largest segment and drives substantial total churn volume.

Synthesis and final recommendation

The final story connects the three methods without confusing prediction and causality.

Integrated answer

EDA	Contract, tenure, and charges are central churn patterns.
Prediction	Random Forest detects more churners; Logistic Regression limits false positives.
Segmentation	Transactional Volatiles is the largest segment; High-Value At-Risk has the highest churn rate.
Causal	Longer contracts and technical support show negative estimated effects on churn.

Recommended business reading

Use prediction to prioritize high-risk customers, segmentation to tailor retention by group, and causal inference to choose interventions worth testing.

Limitations

IBM demo dataset; observational causal assumptions; coefficients and feature importance are predictive, not causal; K-Means depends on encoding and scaling choices.

References

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